

EDEM ROBIN

Network Engineer | Python Developer | Network Automation Enthusiast

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Professional Summary

I am a Network Engineer, Python Developer, and technology enthusiast passionate about designing, automating, and improving modern IT infrastructure.

My expertise sits at the intersection of **network engineering, software development, automation, and systems administration**. I specialize in building reliable network environments, monitoring solutions, automation tools, and practical software applications that solve real-world problems.

I have experience working with ISP infrastructure, enterprise networks, Linux systems, virtualization platforms, monitoring solutions, and network automation technologies.

My goal is to bridge the gap between traditional network engineering and modern infrastructure automation by creating solutions that are scalable, efficient, and easy to operate.

Technical Skills

Networking

- Enterprise Network Design
- ISP Network Support
- Routing and Switching
- TCP/IP Networking
- VLAN Design and Segmentation
- Network Troubleshooting
- Wireless Networking
- VPN Technologies
- Firewall Configuration
- Network Documentation

Networking Technologies

- MikroTik RouterOS
 - Cisco IOS / IOS XE
 - Cisco ASR Platforms
 - BGP Fundamentals
 - OSPF
 - DHCP
 - DNS
 - NAT
 - SNMP
 - IPv4 / IPv6 Fundamentals
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Network Automation

- Python Network Automation
 - pyATS
 - Netmiko
 - Paramiko
 - Ansible
 - Bash Scripting
 - Configuration Automation
 - Network Testing
 - Infrastructure Documentation
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Systems & Infrastructure

- Linux Administration
 - Ubuntu Server
 - Proxmox VE
 - Virtual Machines
 - LXC Containers
 - Docker
 - Nginx
 - Git
 - Self-hosted Infrastructure
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Monitoring & Observability

- Zabbix
 - PRTG
 - Grafana
 - SNMP Monitoring
 - Infrastructure Alerting
 - Performance Analysis
 - Network Visibility Solutions
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Software Development

Programming Languages

- Python
 - JavaScript
 - HTML
 - CSS
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- Bash

Frameworks & Tools

- Django
- FastAPI
- Tailwind CSS
- Android Development
- Git

Databases

- SQLite
- MySQL
- PostgreSQL Fundamentals

Professional Experience

Network Infrastructure & Deployment Engineer

MISMIS Concept Full-time

Sep 2023 Present

Accra, Greater Accra Region, Ghana

Responsibilities:

- Lead and support network infrastructure design and deployment operations.
- Design, support, and maintain reliable networking solutions with custom routing/switching configurations.
- Trouble-shoot complex connectivity and infrastructure issues.
- Support deployment and maintenance of scalable active network infrastructure.

Network Support Engineer

Dimension Data Ghana Full-time

Jun 2022 Present

Ghana

Responsibilities:

- Administered enterprise networking environments utilizing MPLS VPN, Border Gateway Protocol (BGP), and active routing.
- Monitored network infrastructure to optimize uptime and response times.
- Assisted with troubleshooting active connectivity issues.
- Supported community WiFi infrastructure projects.

Project Team Lead

I-Project leg Part-time

Aug 2019 Present

Ghana, Accra

Responsibilities:

- Served as Project Lead for the **DNB APP**, a custom system that allows students to receive their timetable on time, download past questions, and access learning resources for free.
 - Engineered and released the Android APP available publicly on the Google Play Store.
 - Managed user experience, feature updates, and customer satisfaction metrics.
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Android Developer

Freelance

Oct 2018 Present

Ghana

Responsibilities:

- Designed and engineered customized mobile applications on the Android platform.
 - Managed lifecycle, APIs, data flows, and lightweight databases to deliver responsive applications.
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Network Support Engineer

Digital Arena Ghana Full-time

Jan 2020 Oct 2022

East Cantonment, Accra

Responsibilities:

- Installed and maintained customer network equipment.
 - Configured routers, switches, and wireless devices.
 - Provided high-touch technical support and troubleshooting.
 - Assisted with network deployments and customer connectivity solutions.
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Global Service Center (GSC) Representative & Network Engineer

Internet Solutions

Dec 2014 Aug 2019

Responsibilities:

- Provided First Level and Second Level Network Troubleshooting.
 - Developed and built an NMS Interface for active network monitoring.
 - Tracked network events, generated alarms, and resolved service incidents.
 - Managed customer satisfaction and SLA compliance.
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Featured Projects

Enterprise Home Lab Infrastructure

A personal enterprise-style infrastructure environment built for testing, learning, automation, and real-world simulation.

Technologies

- Proxmox VE
- MikroTik Routers
- Linux Servers
- Docker
- Zabbix
- NetBox
- Grafana
- Jellyfin
- NetBird

Implementations

- Designed VLAN-based network segmentation.
- Created dedicated server, management, storage, and lab networks.
- Virtualized multiple services using Proxmox.
- Implemented monitoring and infrastructure documentation.
- Built self-hosted services for learning and experimentation.

NetBox Infrastructure Documentation Platform

Deployed NetBox as a network source of truth for infrastructure management.

Features

- IP Address Management (IPAM)
- Device Inventory
- Network Documentation
- Infrastructure Modeling
- Network Planning

Technologies

- Python
- PostgreSQL
- Nginx
- Linux

TimeGate Attendance System

A web-based attendance management system developed using Django.

Features

- User management
- Attendance tracking
- Database integration
- Web-based interface

Technologies

- Python
 - Django
 - SQLite
 - Tailwind CSS
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Network IP Management System

A FastAPI-based application designed to simplify IP address management.

Features

- IP record management
- API-based architecture
- Lightweight database storage
- Web interface integration

Technologies

- Python
 - FastAPI
 - SQLite
 - Tailwind CSS
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CONNECT56

A Linux networking utility project focused on simplifying connectivity workflows.

Technologies

- Bash scripting
 - Linux networking tools
 - Automation concepts
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Virtualization & Lab Experience

Hands-on experience building and managing virtual environments using:

- Proxmox VE
- VMware Workstation
- ESXi
- GNS3
- EVE-NG
- PNETLab

Used for:

- Network simulations
 - Infrastructure testing
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- Automation development
 - Service deployment
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Education & Certifications

University of Ghana

Bachelor of Science Degree

Udacity

Android Basics Nanodegree

Focus areas:

- Android application development fundamentals
 - Java/Kotlin programming concepts
 - Android UI development
 - Application lifecycle and components
 - Mobile application design principles
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City & Guilds of London

Level 3 Telecommunication Technician Qualification

Focus areas:

- Telecommunications fundamentals
 - Communication systems
 - Network infrastructure concepts
 - Cabling and connectivity principles
 - Installation and troubleshooting practices
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Current Learning Focus

Currently developing deeper expertise in:

- Network Automation
 - Python for Network Engineers
 - Cisco pyATS Testing Framework
 - Ansible Automation
 - Advanced Routing Technologies
 - Infrastructure as Code
 - DevOps Practices
 - Cloud Infrastructure
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Professional Interests

- Network Automation
 - Infrastructure as Code
 - Open Source Networking
 - Enterprise Network Design
 - Software Engineering
 - Linux Systems
 - Cloud Technologies
 - Self-hosted Infrastructure
 - DevOps Engineering
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Career Objective

To become a highly skilled network automation and infrastructure engineer by combining networking expertise with software development skills to build intelligent, scalable, and automated systems.

I am passionate about transforming traditional network operations into modern, programmable, and reliable infrastructure.